



New Results

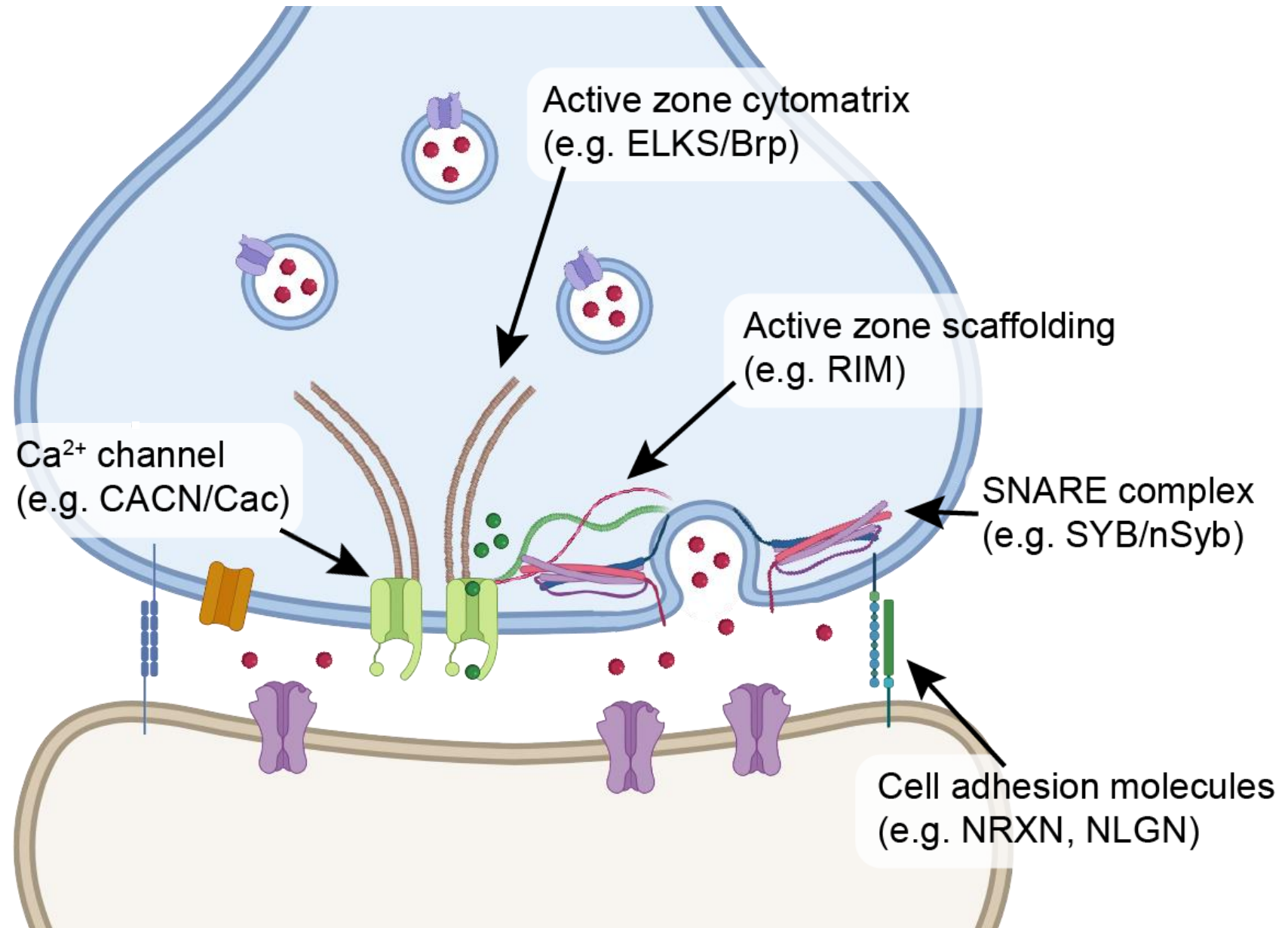
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Deaf1 regulates the coordinated repression of synaptic genes to limit synaptogenesis

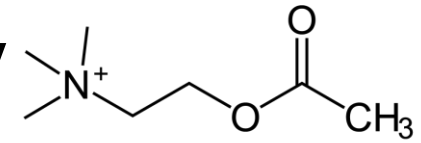
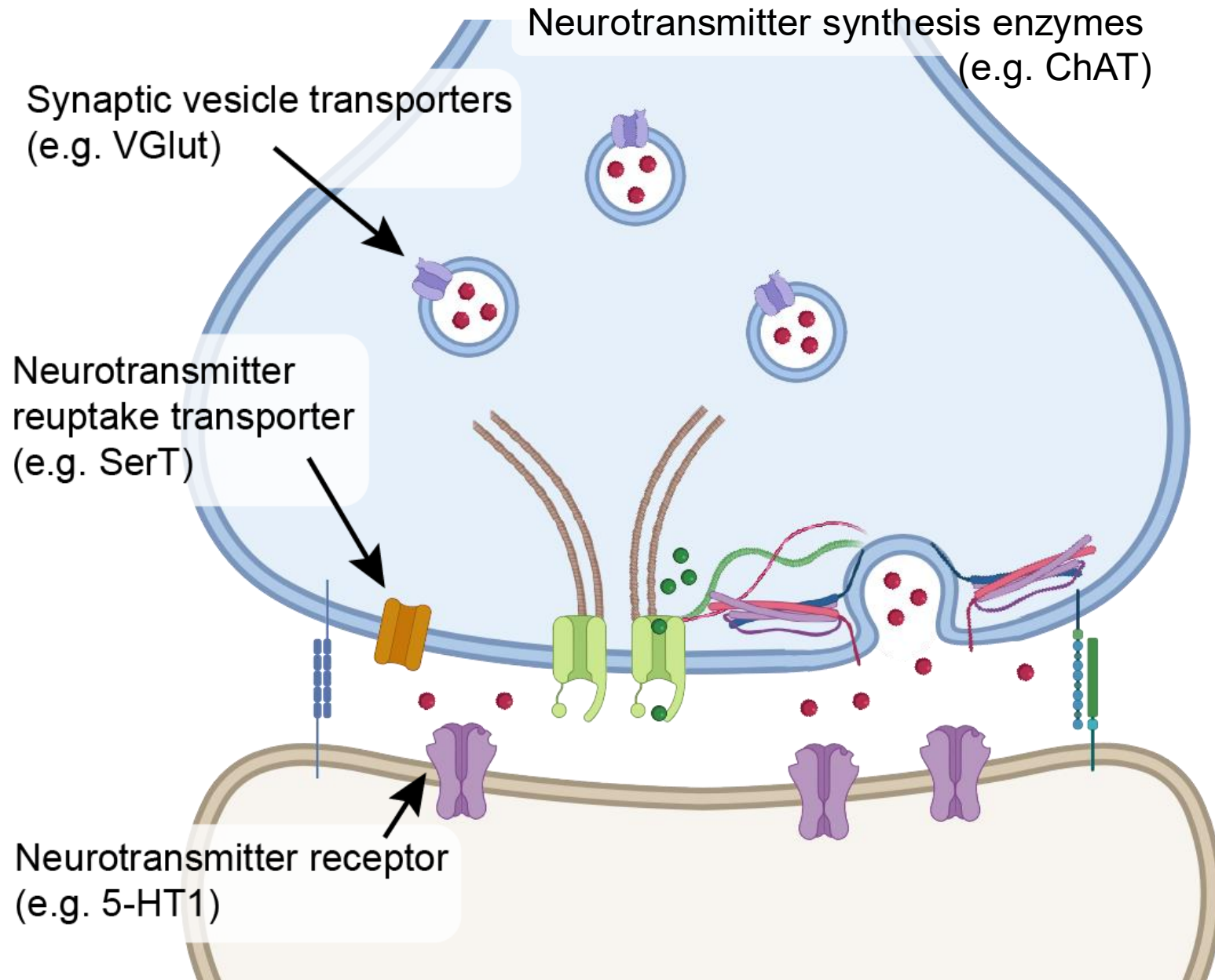
 James A. Kentro,  Gunjan Singh,  Tuan M. Pham,  Justin Currie,  Saniya Khullar,  Audrey T. Medeiros,  Erica Larschan,  Kate M. O'Connor-Giles

doi: <https://doi.org/10.1101/2024.10.30.621128>

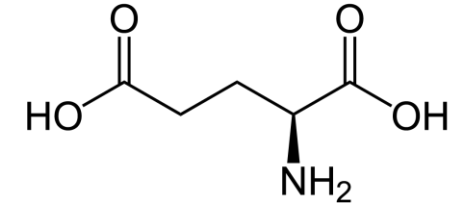
Synapse development and function is reliant on hundreds of components



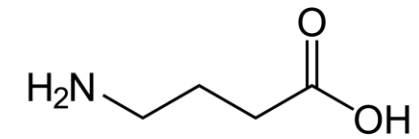
Many synaptic components designate neuronal identity



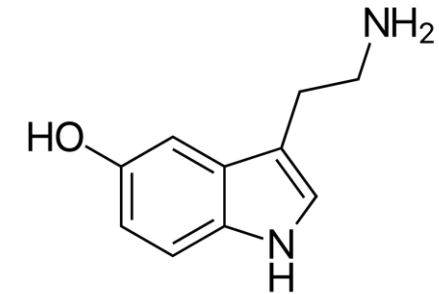
Acetylcholine



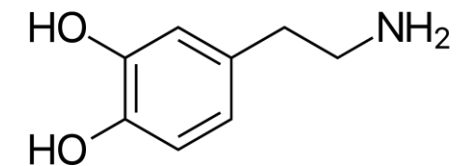
Glutamate



GABA

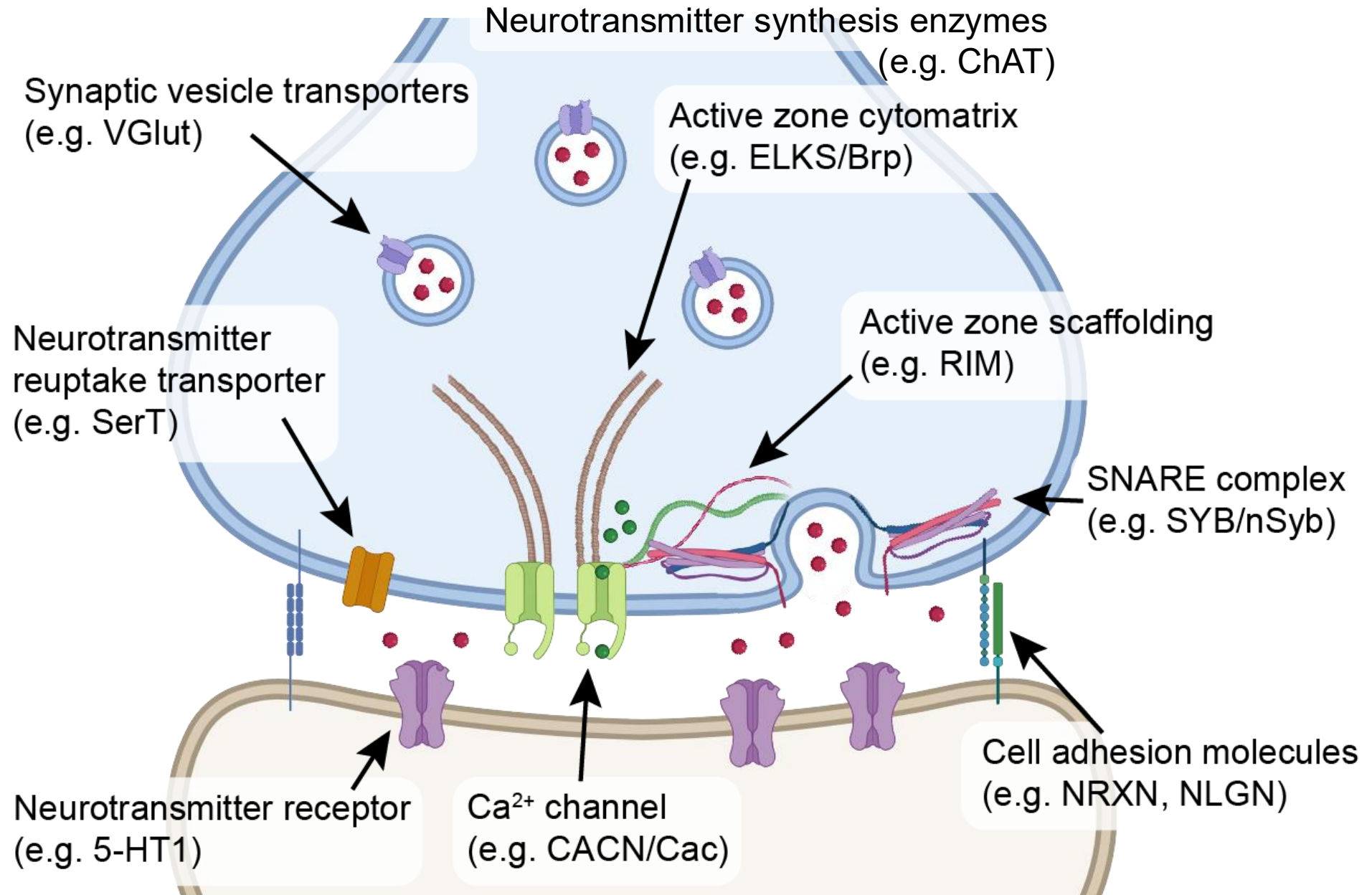


Serotonin



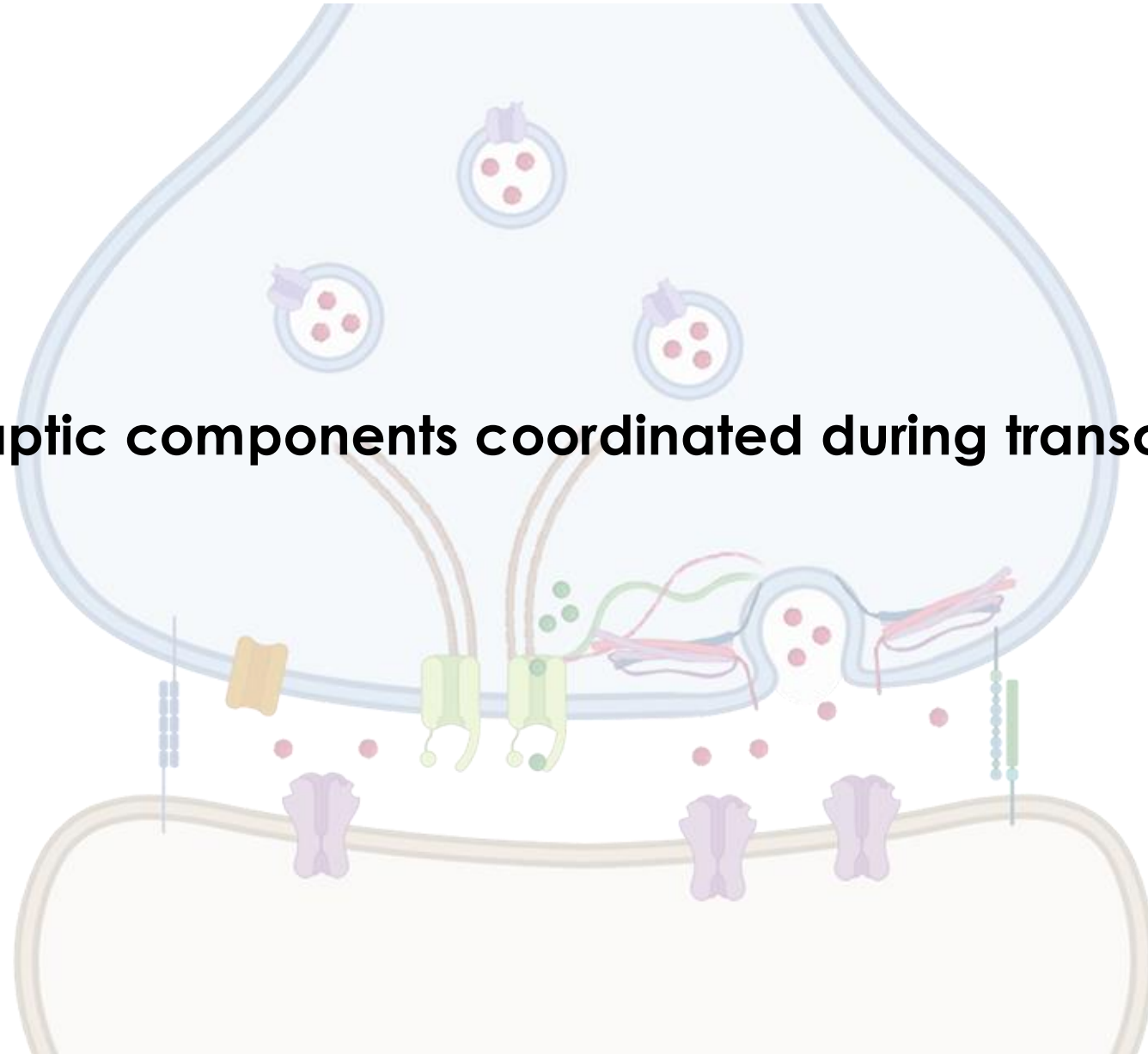
Dopamine

Synaptic components function together to achieve transmission

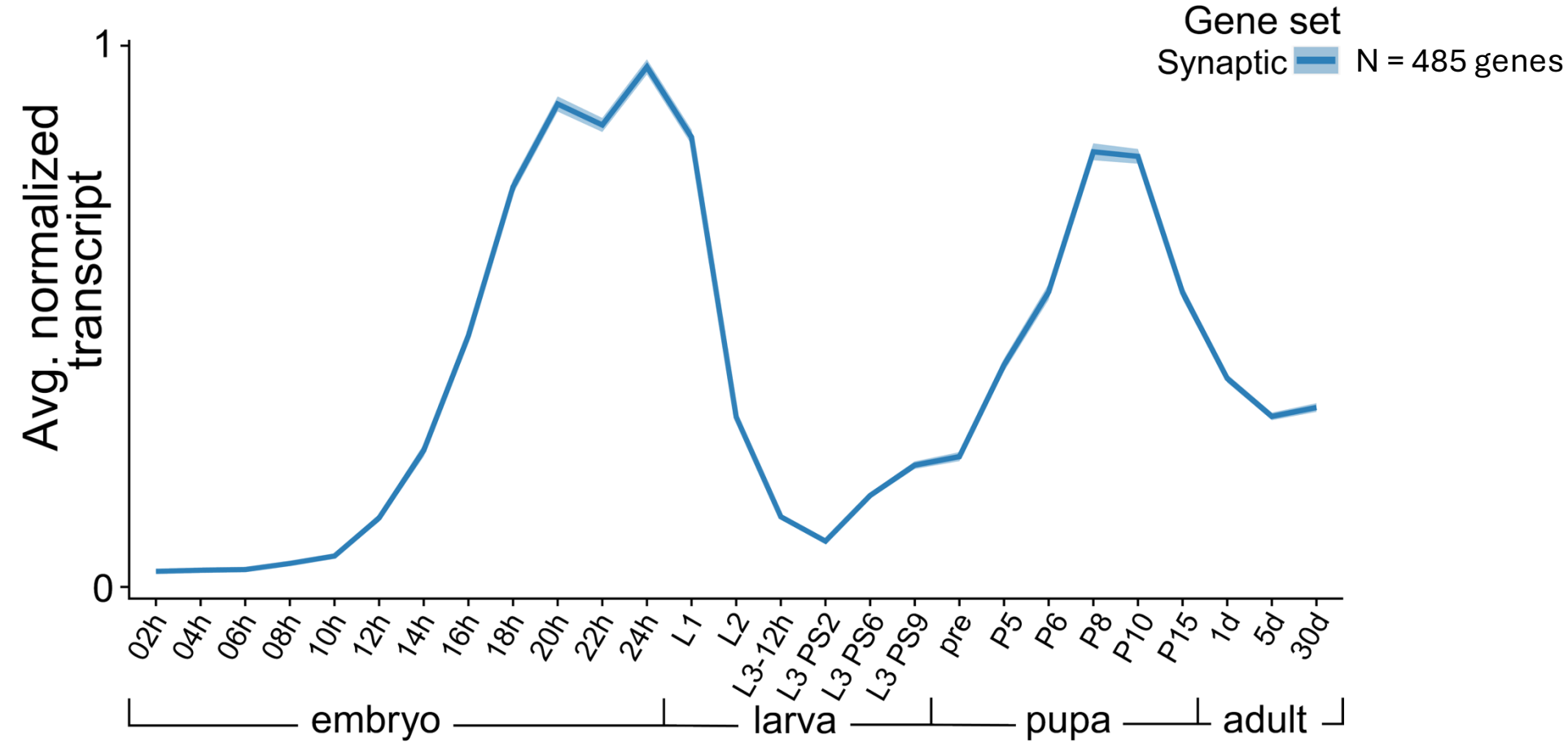


Synaptic components function together to achieve transmission

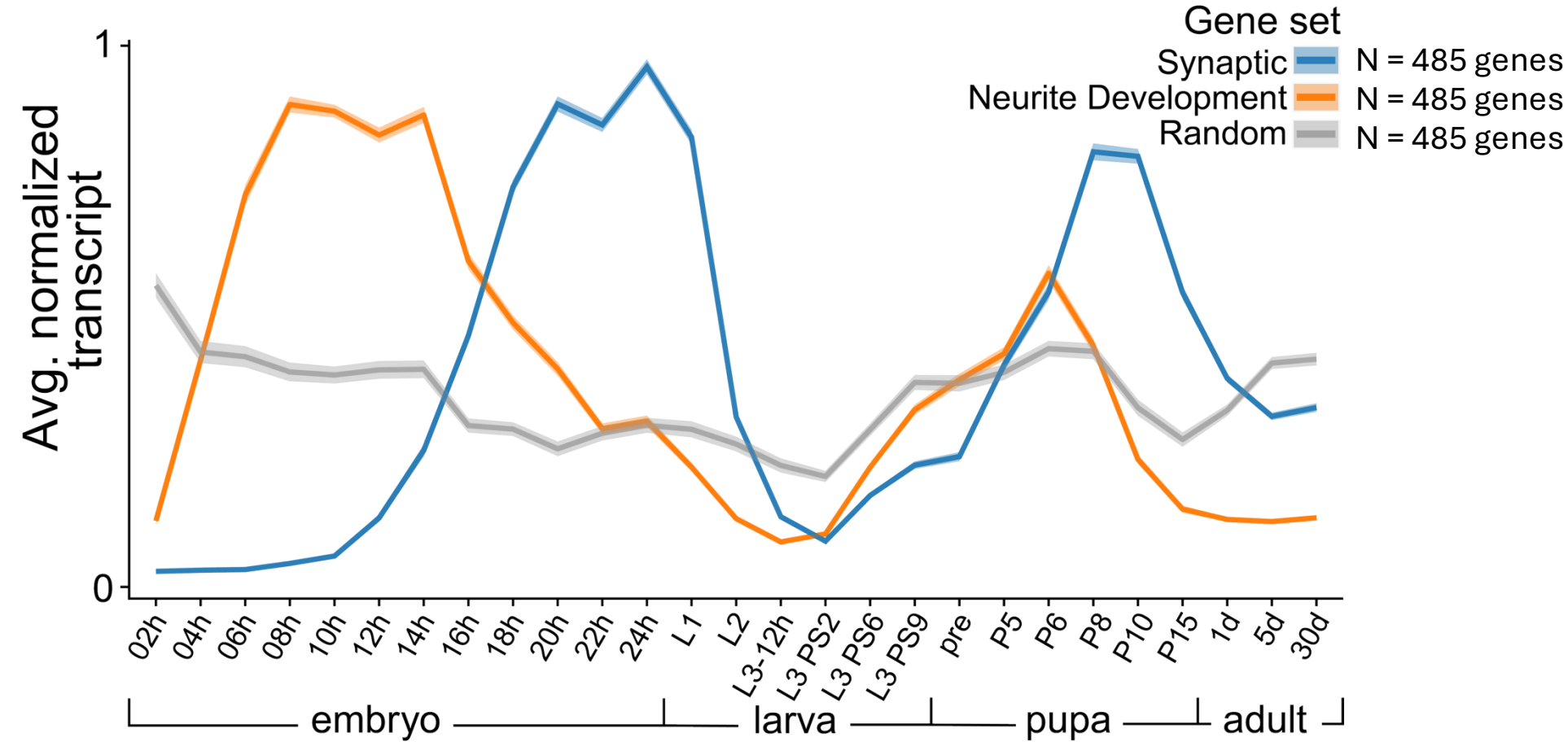
Are synaptic components coordinated during transcription?



Synaptic genes share a temporal pattern of expression

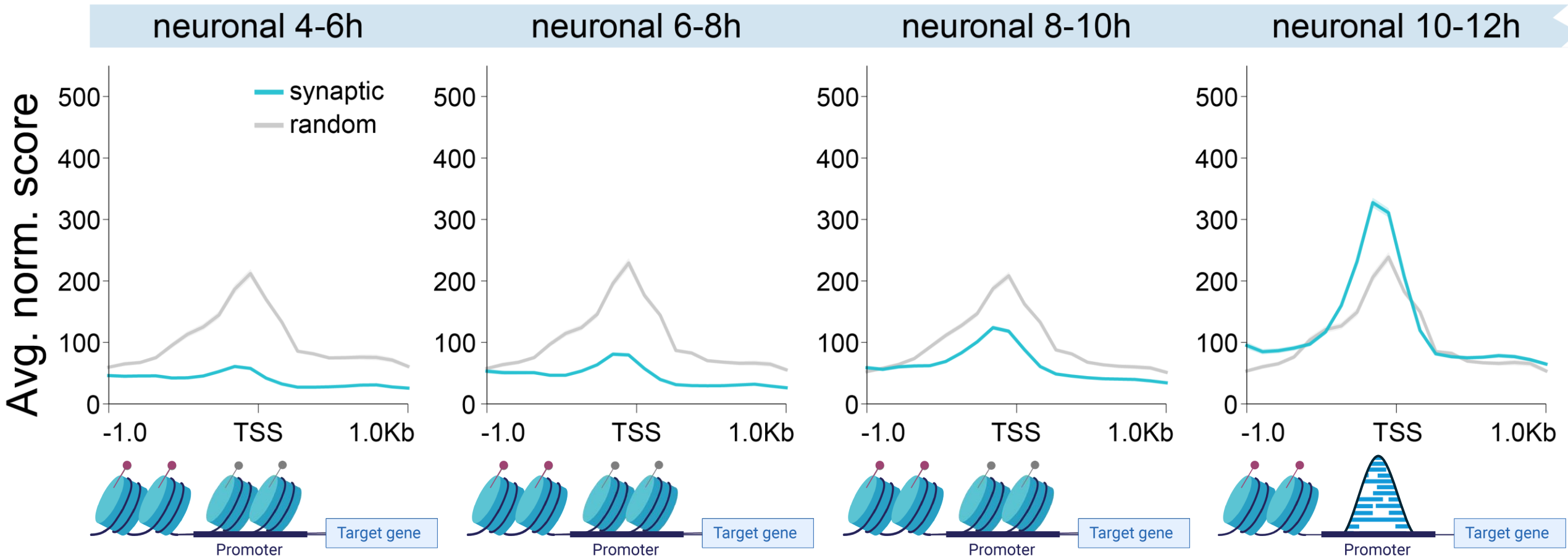


Synaptic genes share a temporal pattern of expression

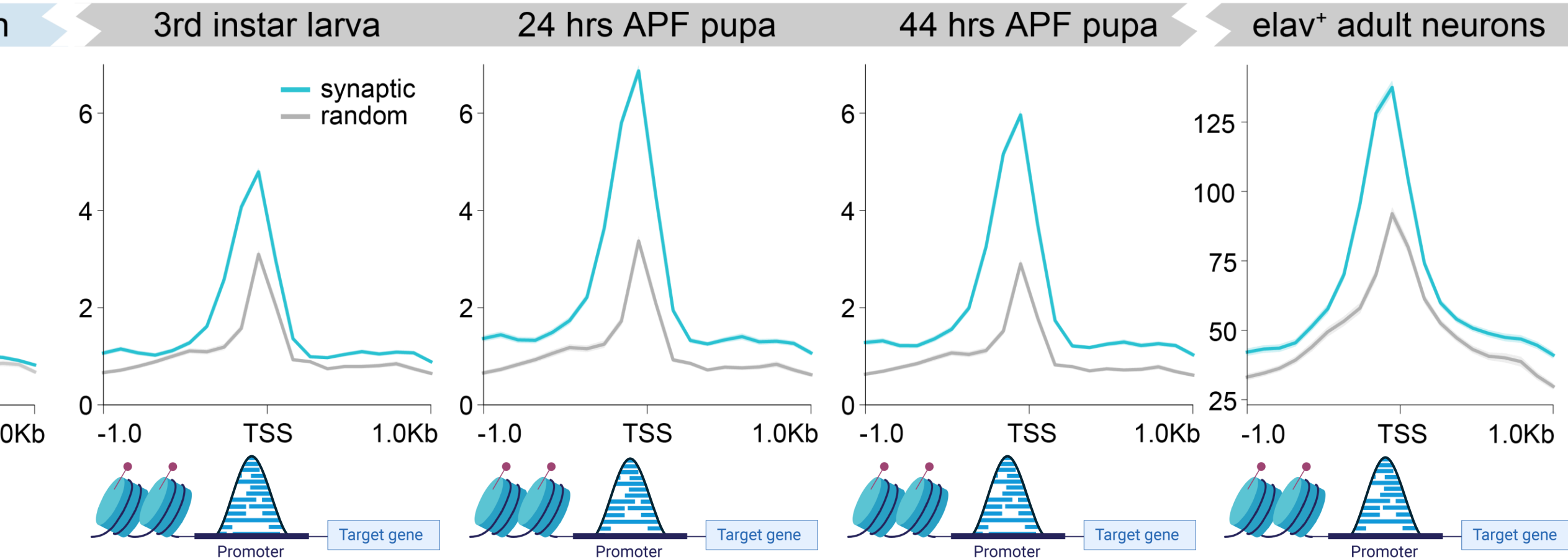


How is coordinated synaptic gene transcription regulated?

Chromatin accessibility is dynamically and coordinately regulated across synaptic genes

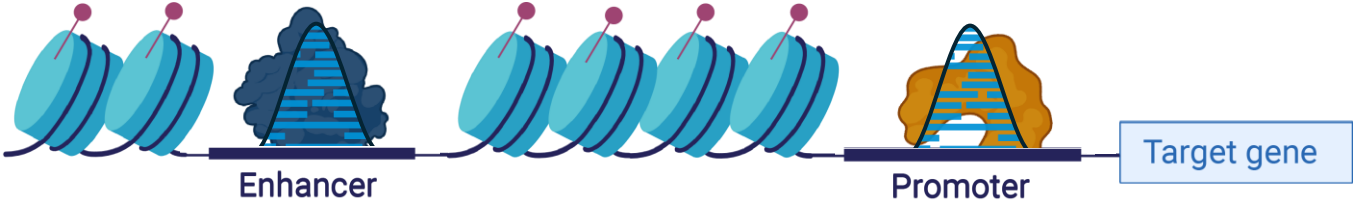


Chromatin accessibility is dynamically and coordinately regulated across synaptic genes

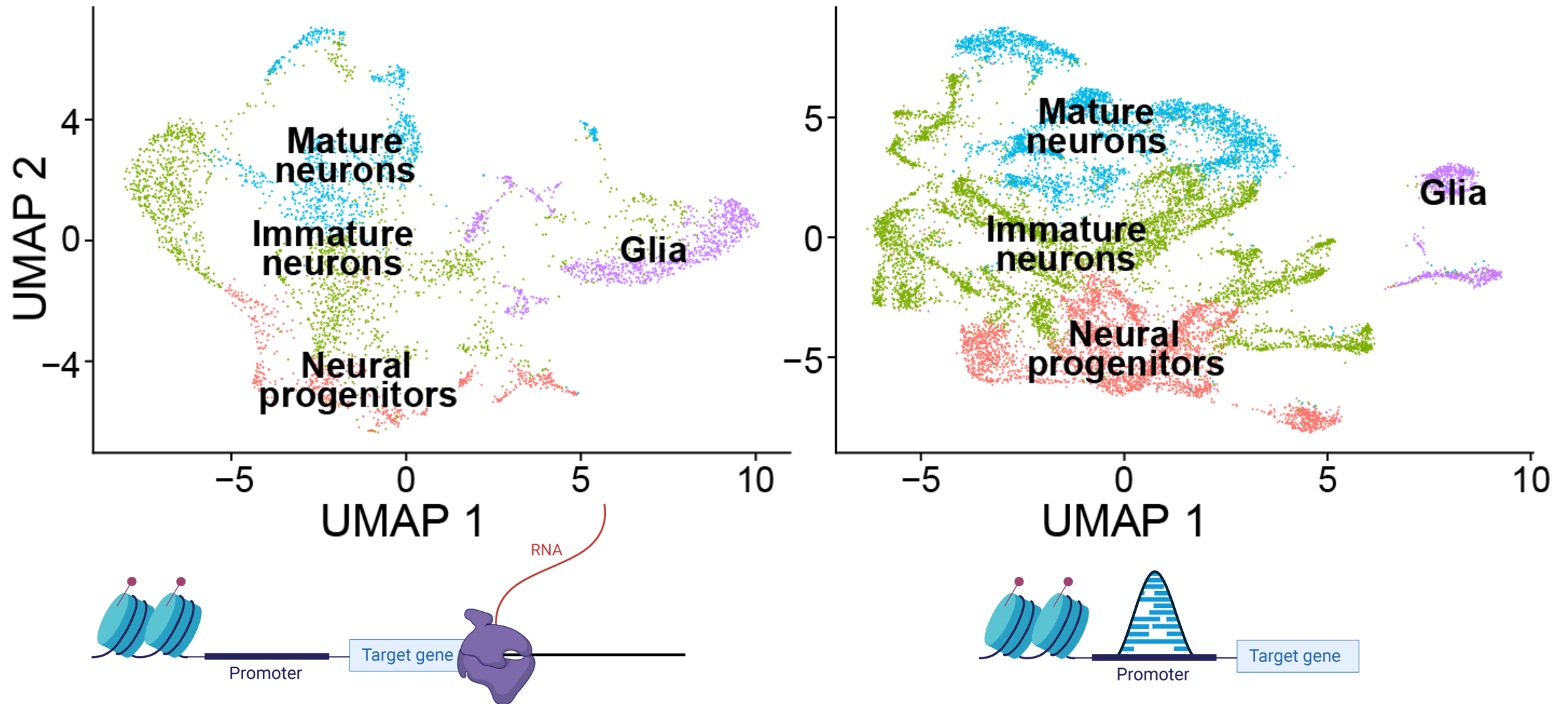


DEAF1 motifs and binding are enriched at dynamic chromatin near synaptic genes

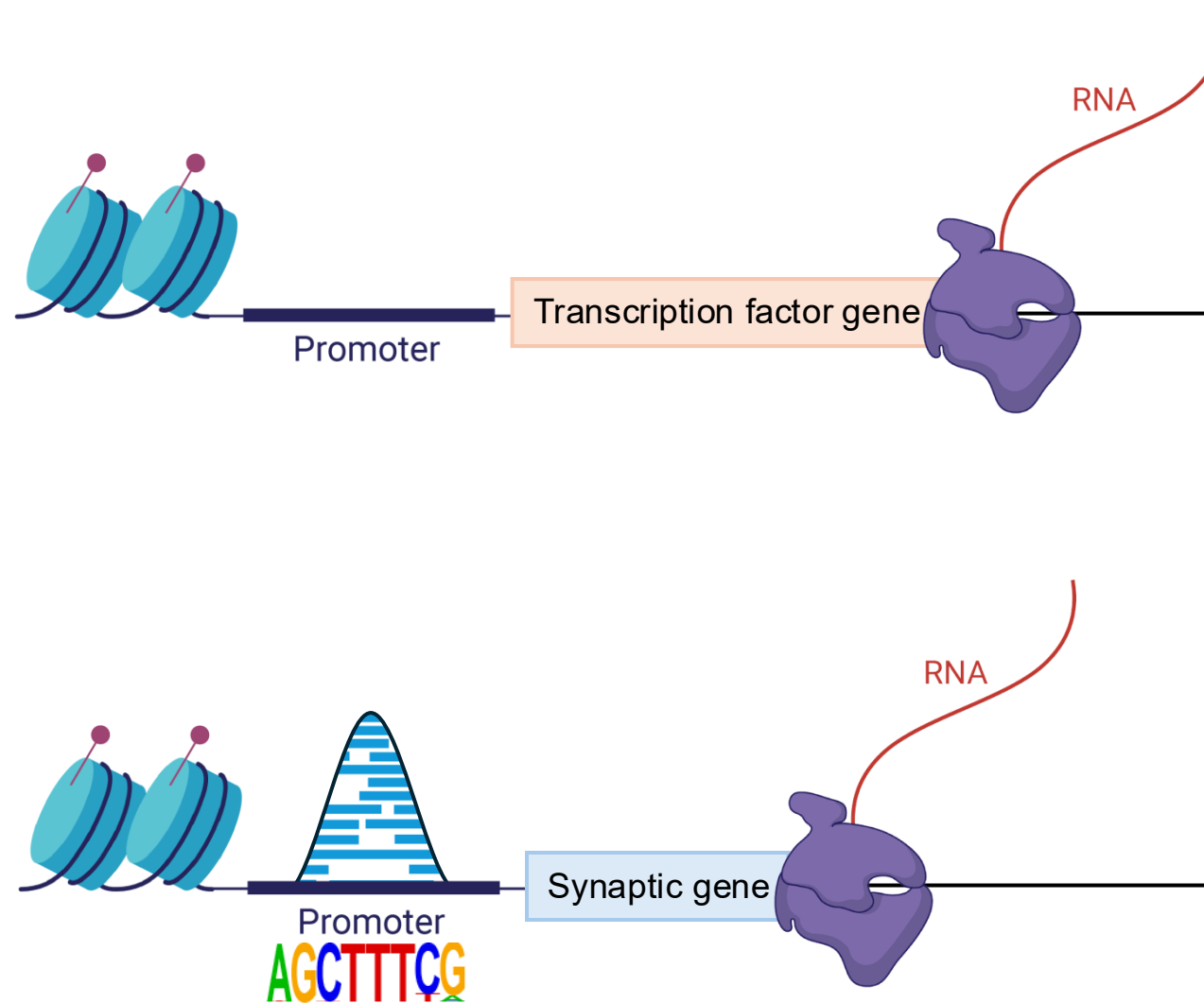
Motif	$\Sigma\text{-log}(p)$	Best match
AGCTTTTCG	553	Deaf1
GAGAGCCA	343	GAF/Clamp
TTTCTTTT	296	pan
CCTGGCCA	239	pho
ATTTTGCA	222	byn
GCAGGC	200	Med
CCACCCCCC	178	run



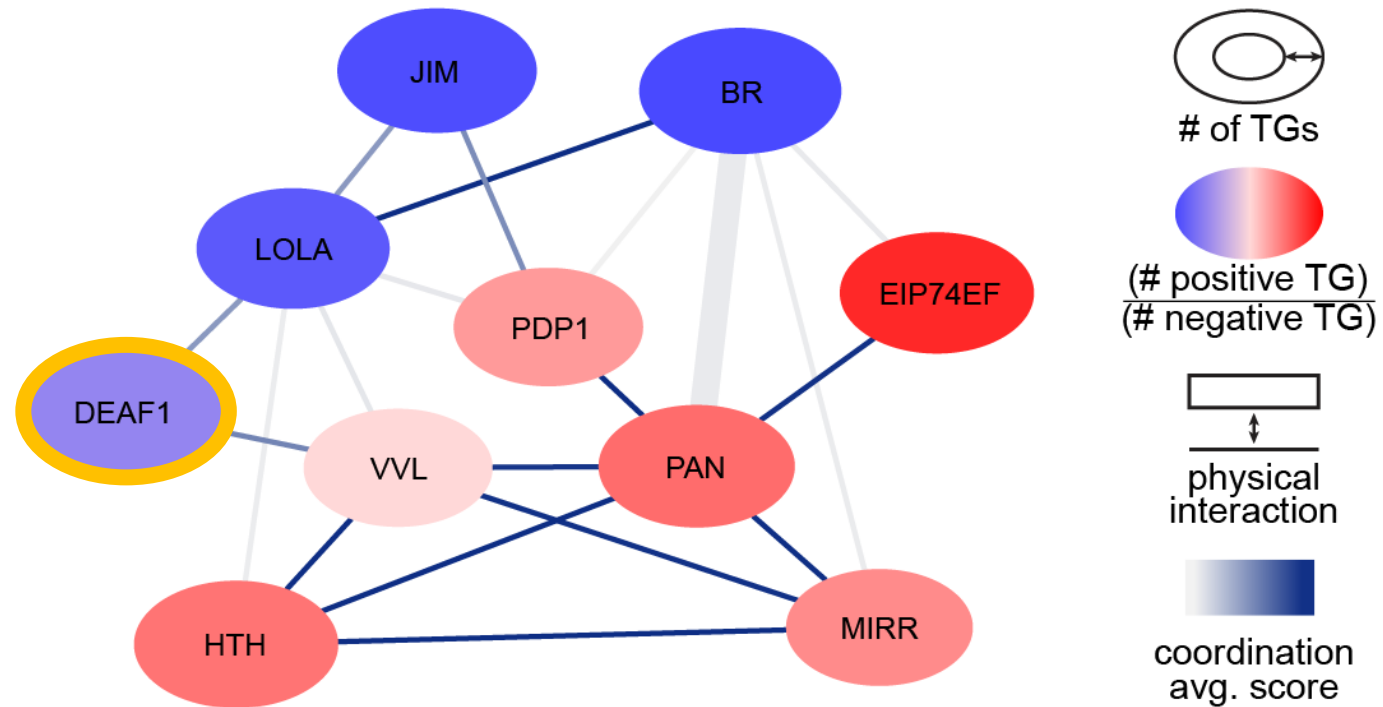
Larval brain single cells annotated by developmental stage



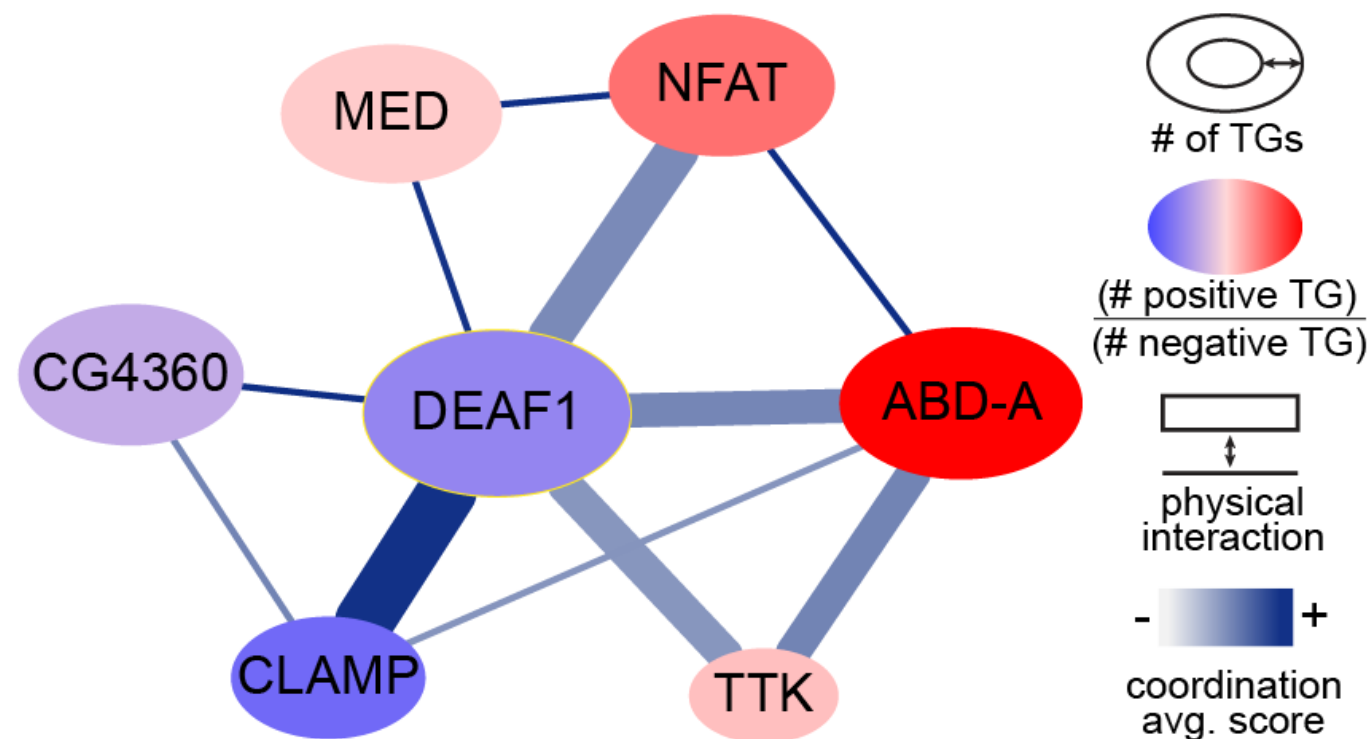
NetREm compares TF with target gene expression and binding motif accessibility



NetREm predicts the top TFs to regulate the most synaptic genes

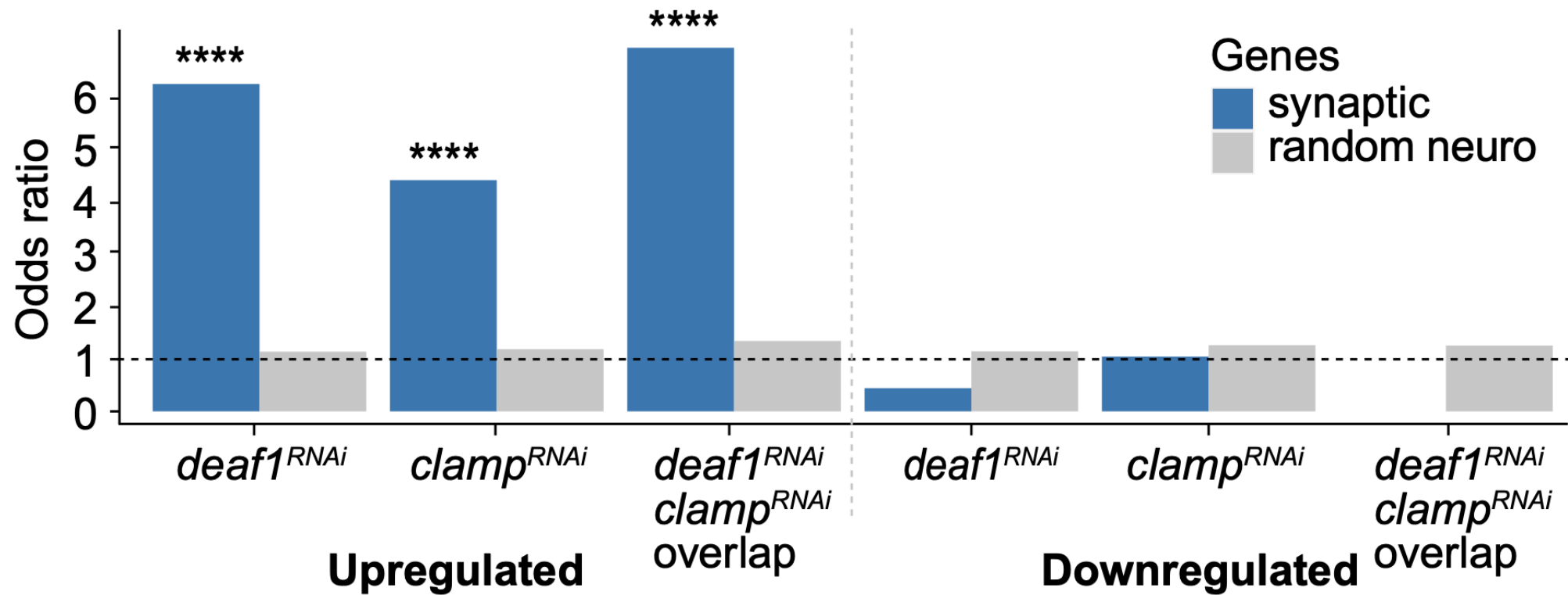


DEAF1 and CLAMP are predicted to co-repress synaptic gene expression



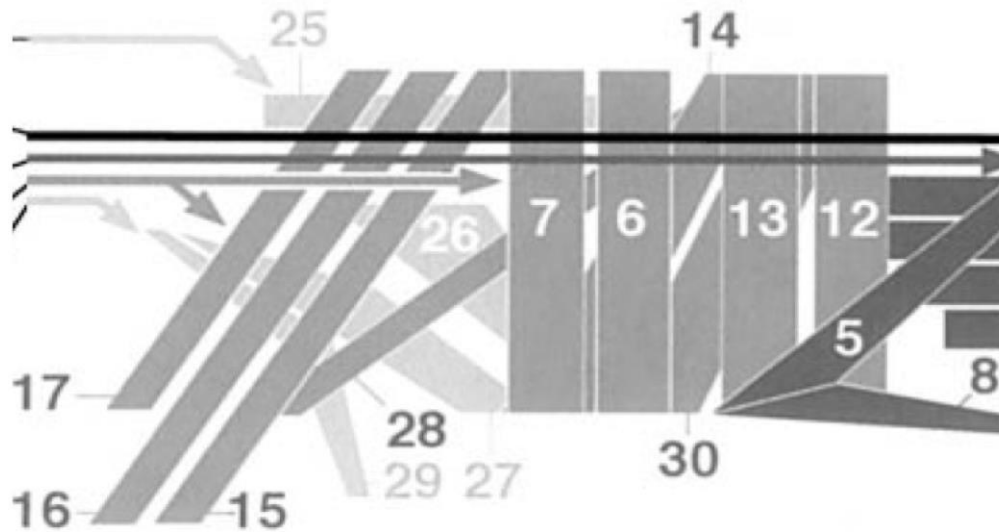
Loss of *deaf1* or *clamp* derepresses synaptic gene expression

Synaptic genes are significantly upregulated

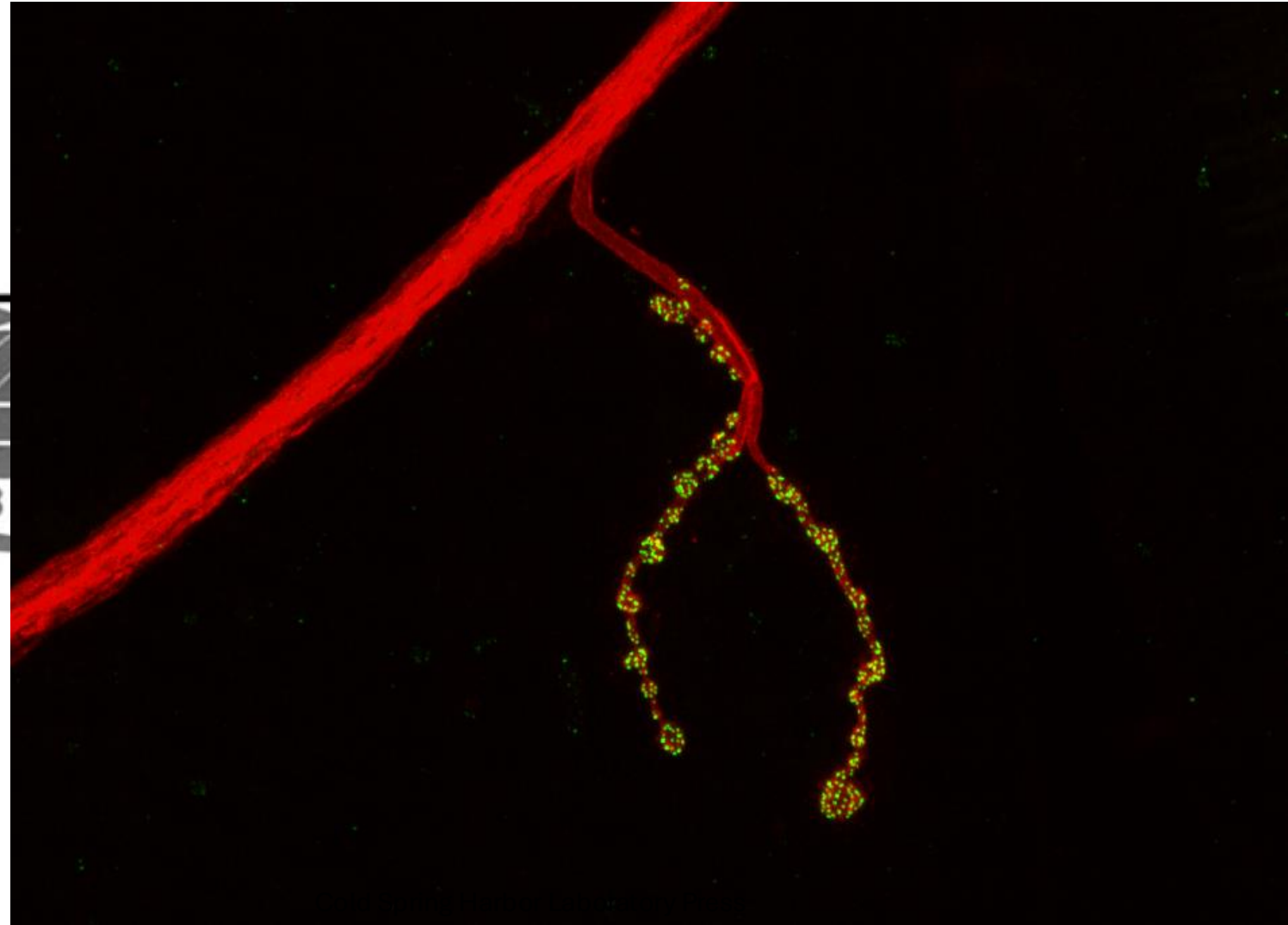


Do DEAF1 and CLAMP regulate synapse formation *in vivo*?

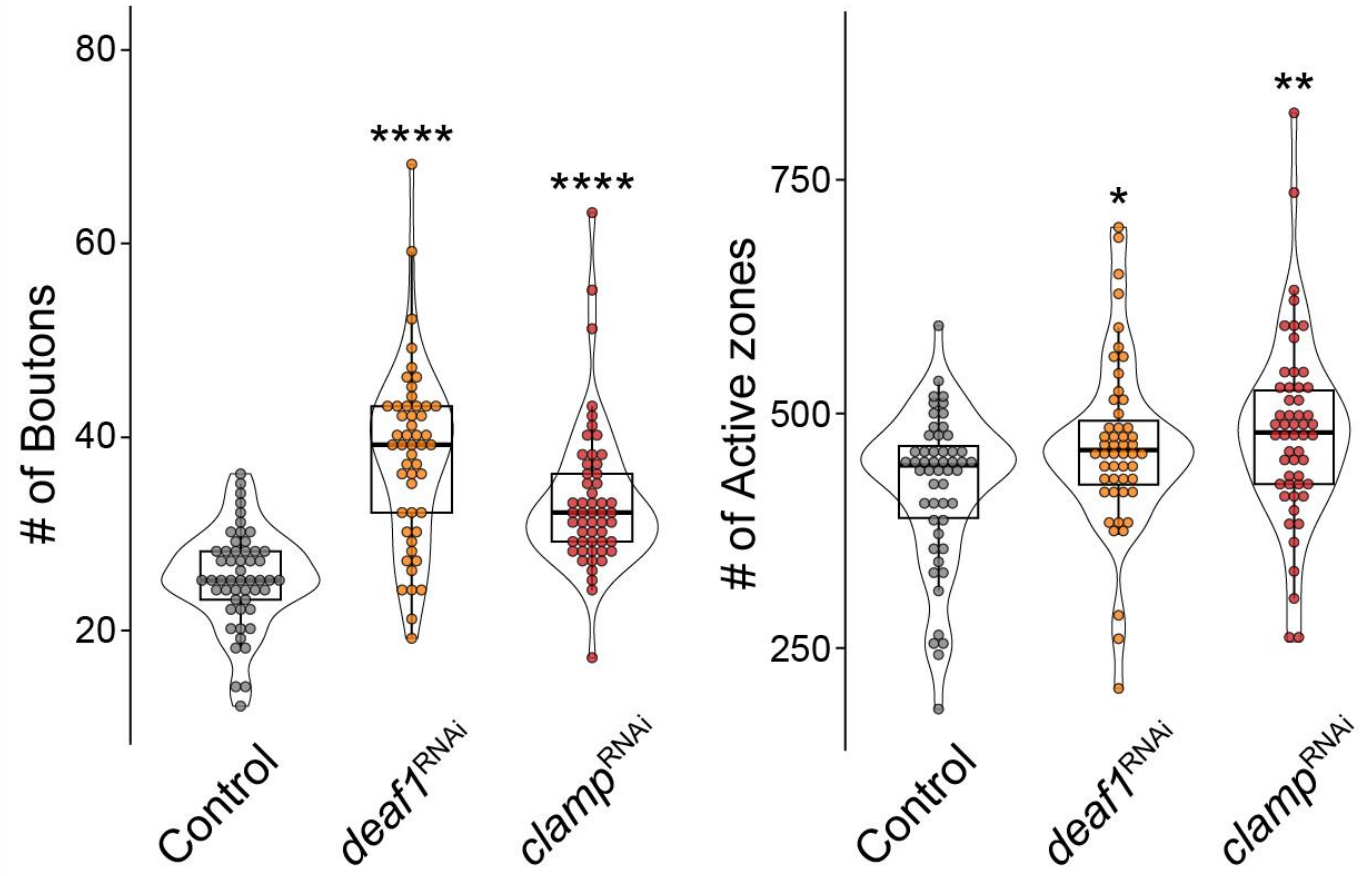
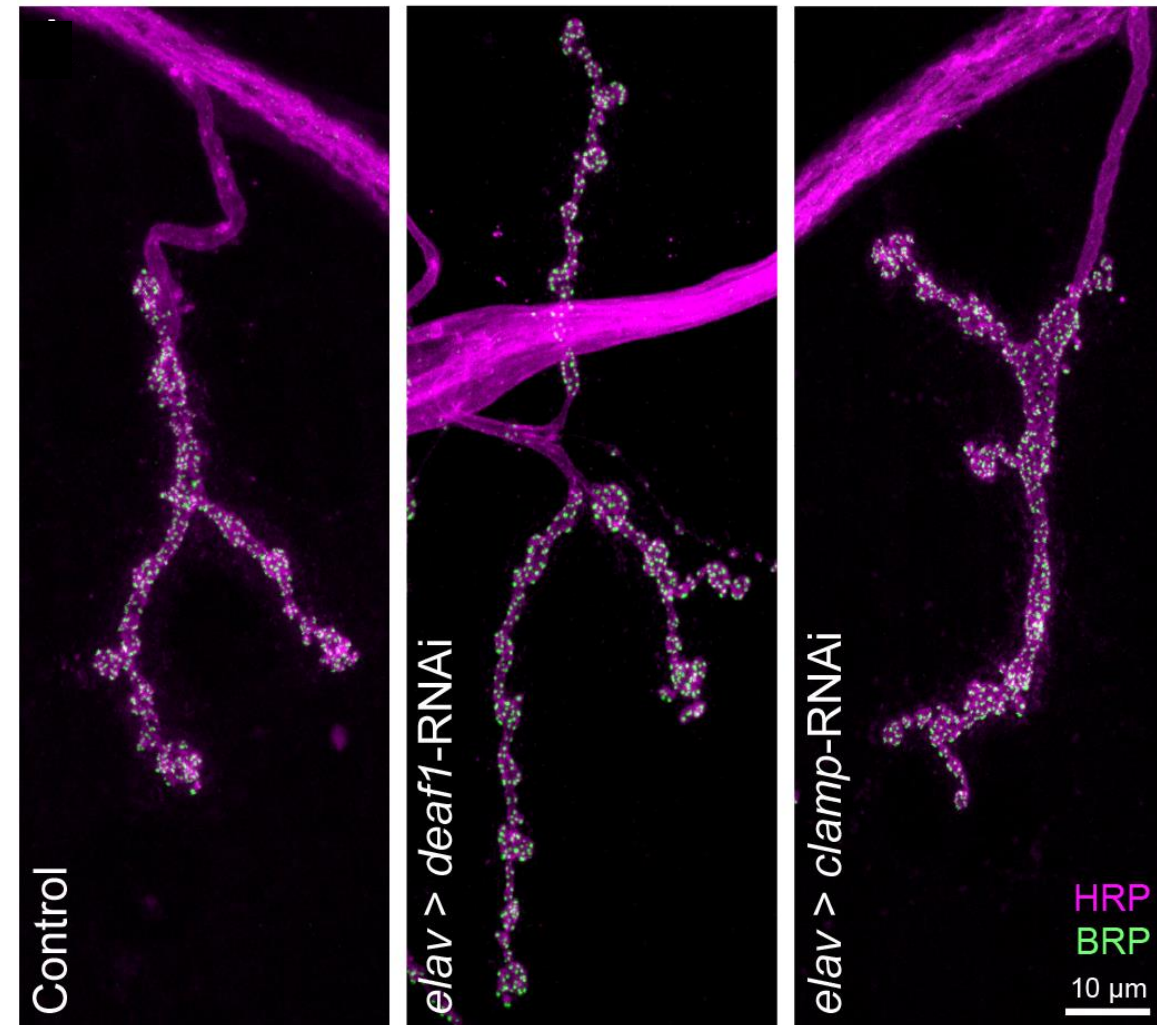
- Assess synaptic growth at larval NMJs
 - Counting synaptic boutons
 - Counting synaptic active zones



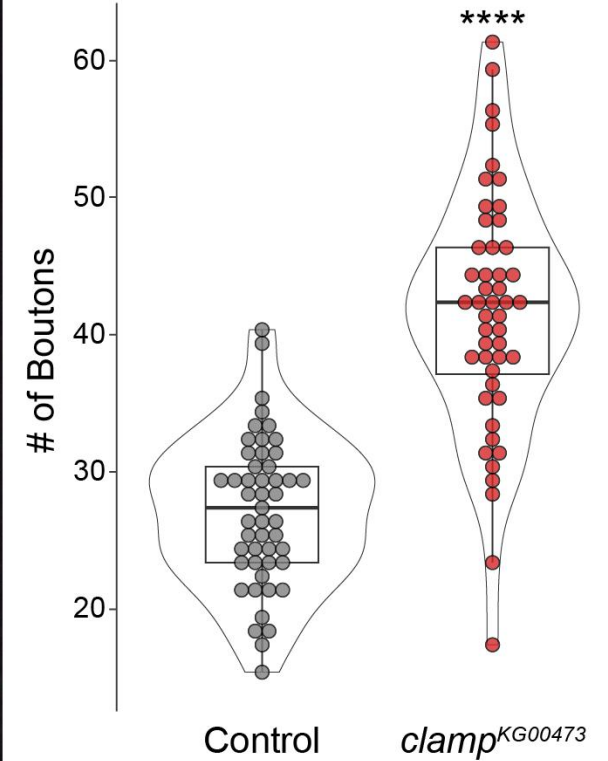
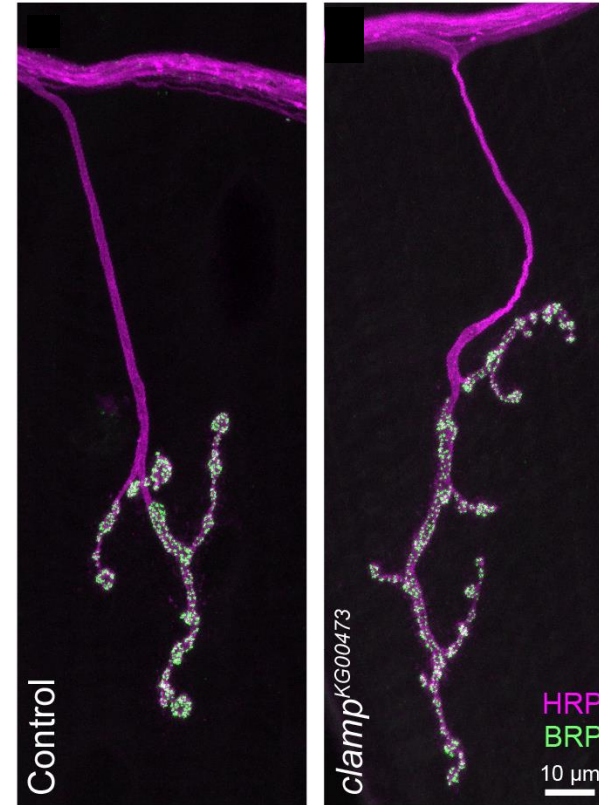
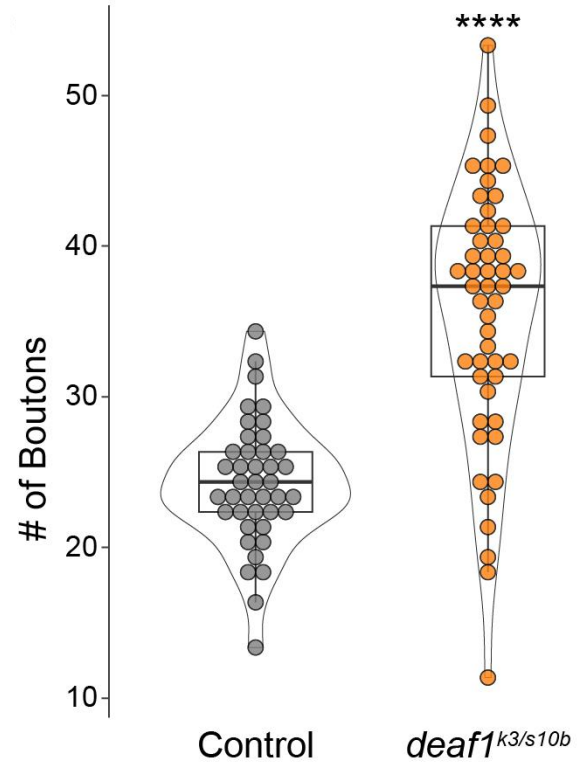
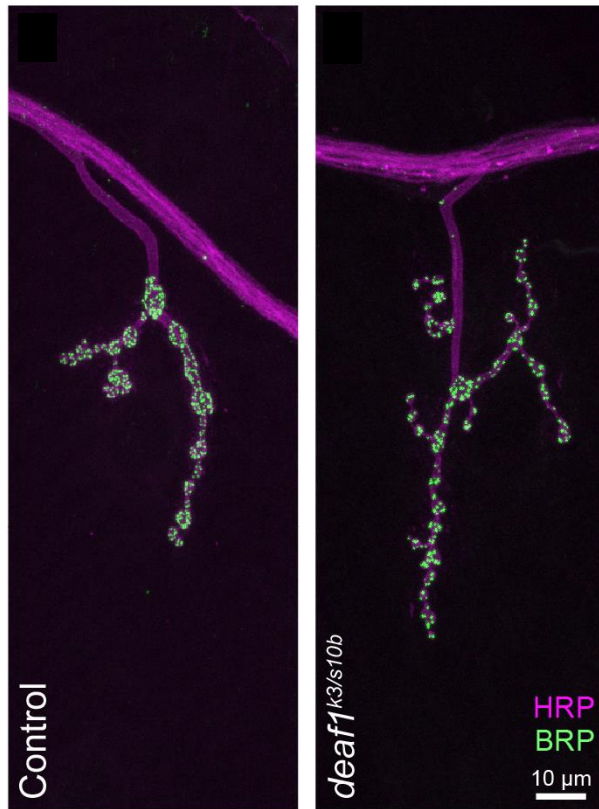
Hoang and Chiba, *Dev. Biology* 2001



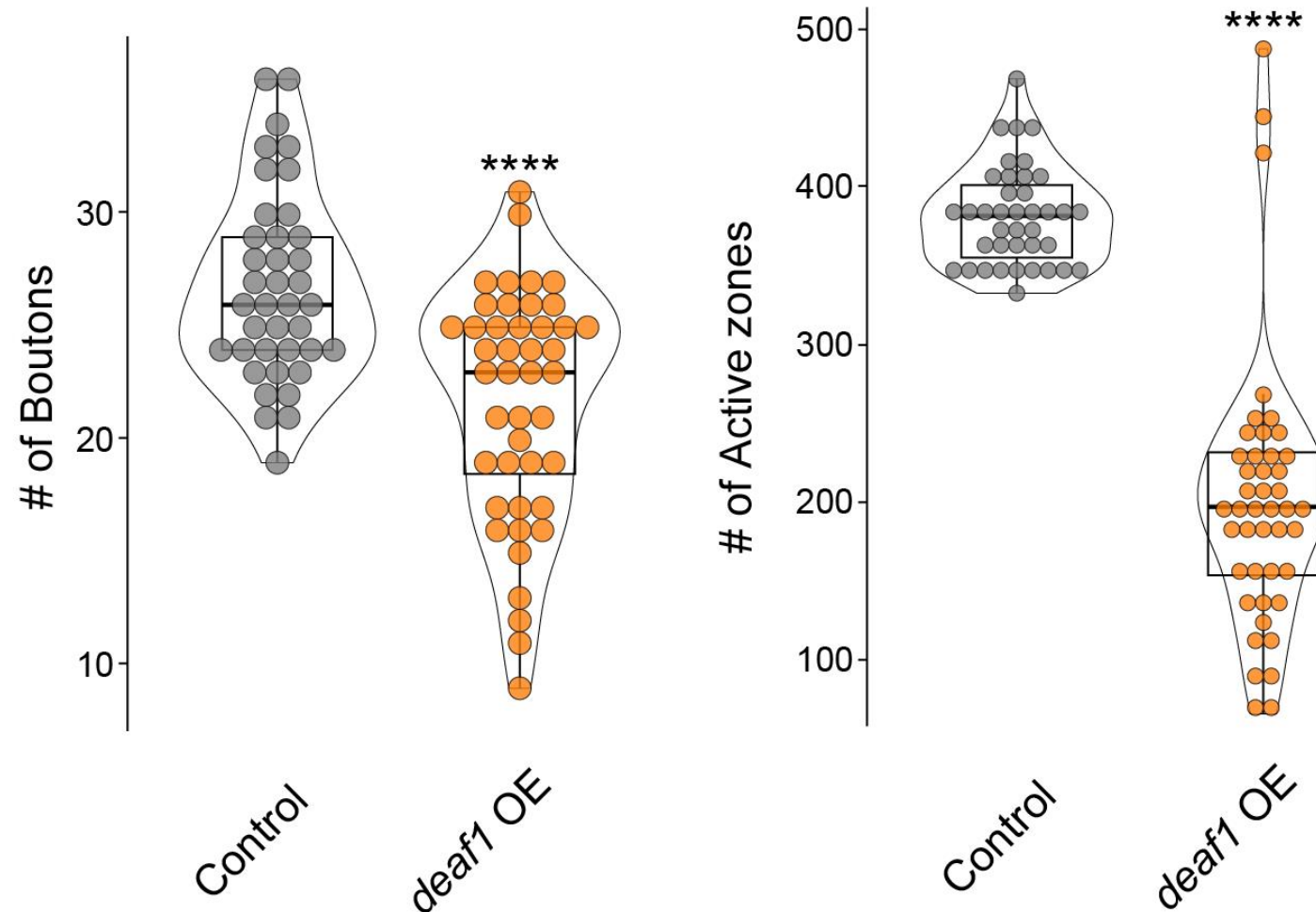
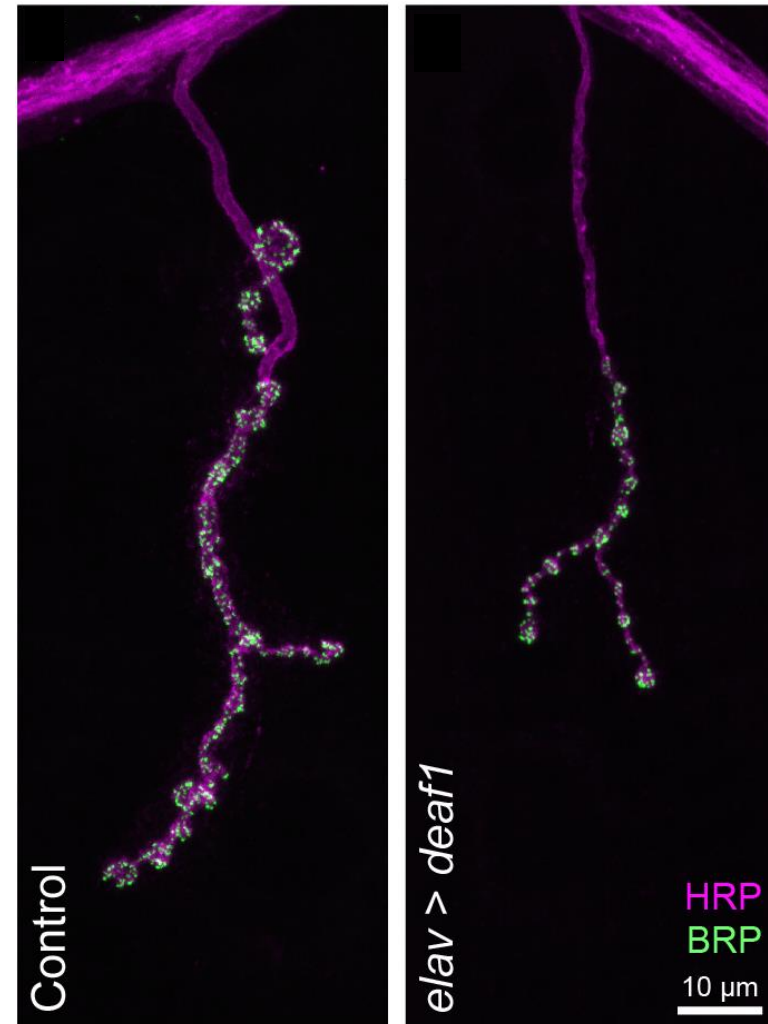
DEAF1 and CLAMP restrain synapse formation



DEAF1 and CLAMP mutations cause dysregulation of synapse formation



DEAF1 misexpression is sufficient to repress synapse formation



Acknowledgements



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